Sleep Health & Lifestyle Intelligence Report

Strategic Analysis

Executive Overview

This report presents a comprehensive analysis of the Sleep Health and Lifestyle Dataset, encompassing 374 individuals across 13 health, lifestyle, and demographic variables. The analysis applies statistical rigor, visual intelligence, and machine learning insights to surface actionable findings for organisational health strategy, workforce wellness investment, and clinical triage optimisation.

Dataset	374 individuals, 13 variables
Analysis Date	February
Prepared By	Shivam Juneja
Classification	Project
Methods Used	EDA, Statistical Testing, ML Classification
ML Accuracy	Up to 91% (Random Forest Classifier)

Key Numbers

374	41.4%	88%	91%
Individuals Analysed *13 variables each*	Have a Sleep Disorder *155 individuals*	Show Elevated Blood Pressure *AHA Classification*	ML Prediction Accuracy *Random Forest*

1. Population Overview

The dataset captures 374 individuals across a range of occupations, age groups, and health profiles. Understanding who is in this population is the foundation for every insight that follows. Of the 374 subjects, 219 (58.6%) present with no sleep disorder. The remaining 155 individuals (41.4%) are clinically classified with either Insomnia (20.6%) or Sleep Apnea (20.9%) - a prevalence rate significantly higher than the global baseline of approximately 30%.

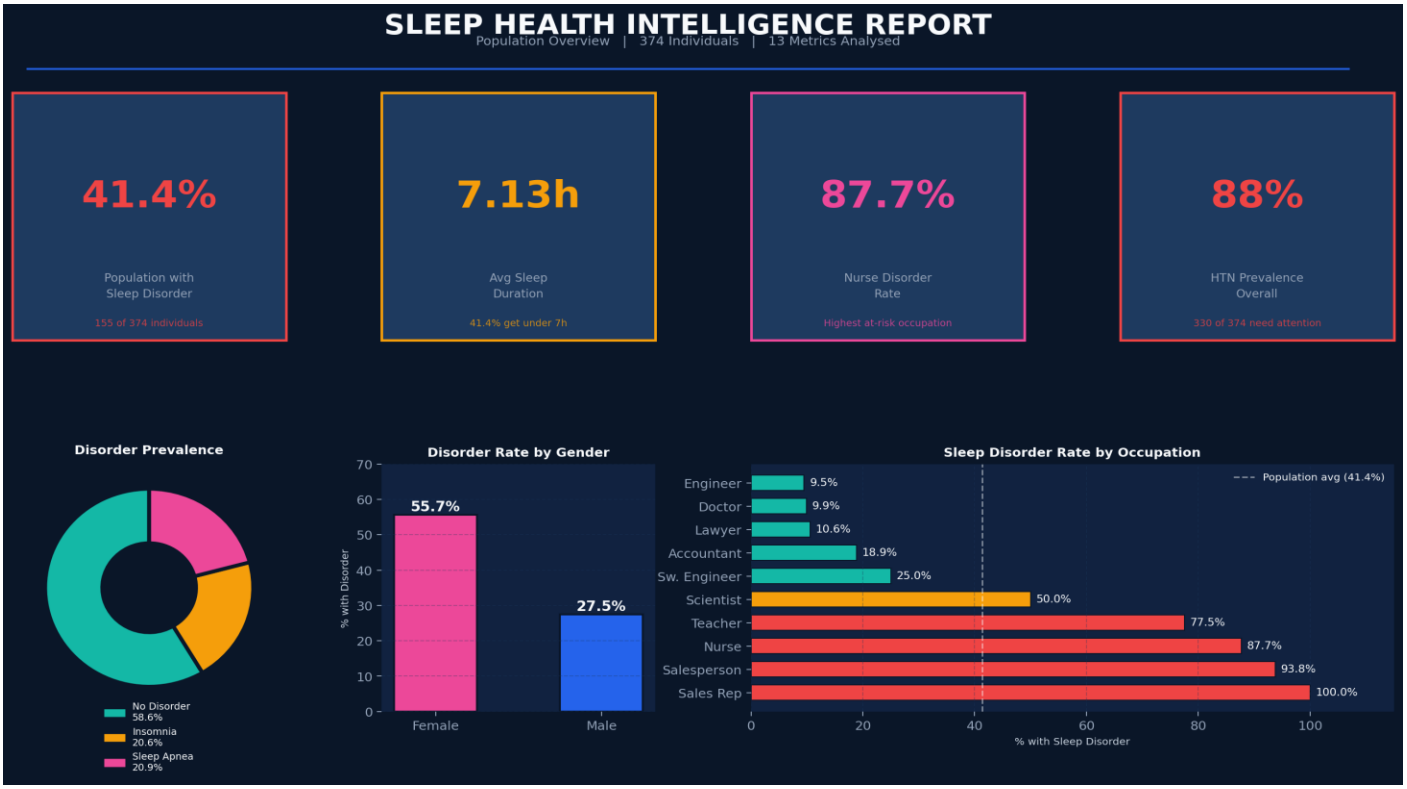

Figure 1: Population overview | disorder prevalence, gender breakdown, and occupation risk rates.

1.1 Disorder Prevalence by Group

Disorder Type	Count	% of Total	Avg Sleep	Sleep Quality	Avg Stress
No Disorder	219	58.6%	7.36h	7.63 / 10	5.11 / 10
Insomnia	77	20.6%	6.59h	6.53 / 10	5.87 / 10
Sleep Apnea	78	20.9%	7.03h	7.21 / 10	5.67 / 10

Key Insight: Gender Disparity is Striking

Female disorder rate: 55.7% vs. Male disorder rate: 27.5%

Women are more than 2x as likely to have a sleep disorder in this population.

Female subjects account for 85.8% of all Sleep Apnea cases (67 of 78), a finding that challenges the clinical stereotype of Sleep Apnea as a predominantly male condition.

2. Health & Wellness Metrics Deep Dive

Beyond simple counts, the power of this dataset lies in the divergence of health metrics across disorder groups. Each of the six metrics below tells a distinct story about what separates disordered from healthy sleepers.

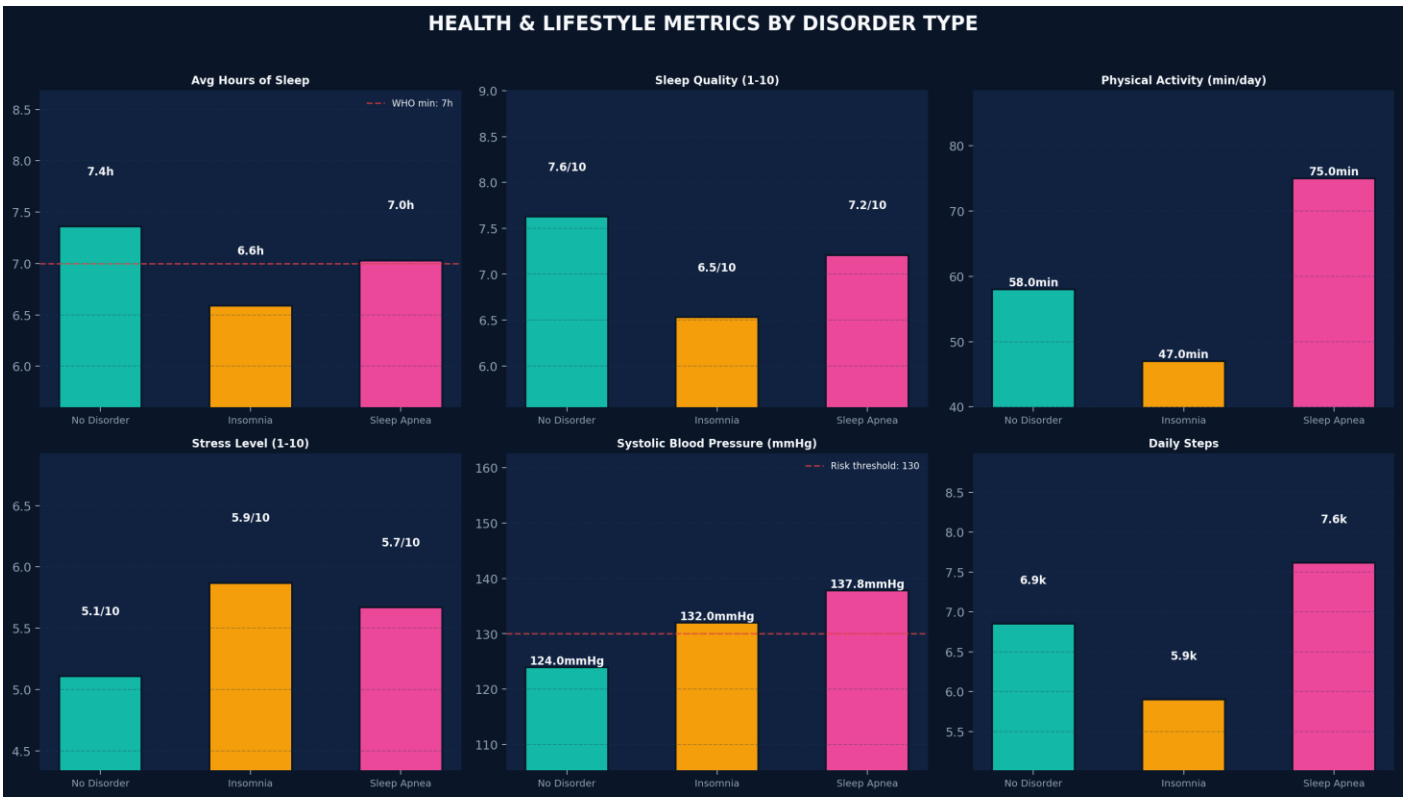

Figure 2: Six key health metrics compared across No Disorder, Insomnia, and Sleep Apnea groups.

2.1 Critical Metric Differentials

Metric	No Disorder	Insomnia	Sleep Apnea
Sleep Duration	7.36h	6.59h	7.03h
Sleep Quality	7.63 / 10	6.53 / 10	7.21 / 10
Physical Activity	58 min/day	47 min/day	**75 min/day**
Stress Level	5.11 / 10	**5.87 / 10**	5.67 / 10
Systolic Blood Pressure	124.0 mmHg	**132.0 mmHg**	**137.8 mmHg**
Daily Steps	6,853	**5,901**	7,619

Analyst Note: Sleep Apnea Presents a Paradox

Sleep Apnea patients are more physically active (75 min/day vs. 58 average) and walk more (7,619 steps vs. 5,901 for Insomnia),
yet carry the highest blood pressure burden (avg systolic: 137.8 mmHg) and the highest Stage 2 Hypertension rate (78.2%).
This suggests Sleep Apnea is a physiological condition independent of lifestyle effort and demands clinical intervention, not just lifestyle coaching.

2.2 WHO Sleep Adequacy Analysis

The World Health Organization recommends 7-9 hours of sleep for adults. Only 58.6% of this population meets that standard. The gap is most extreme for Insomnia sufferers, where only 10.4% achieve adequate sleep duration.

Group	Achieving 7-9h (WHO Adequate)	Below 7h (Insufficient)
No Disorder	79.0%	21.0%
Insomnia	**10.4%**	**89.6%**
Sleep Apnea	48.7%	51.3%

3. Cardiovascular & BMI Risk Matrix

Sleep disorders do not exist in isolation. They intersect with body composition and cardiovascular health in ways that have profound implications for long-term health costs, employee productivity, and organisational liability.

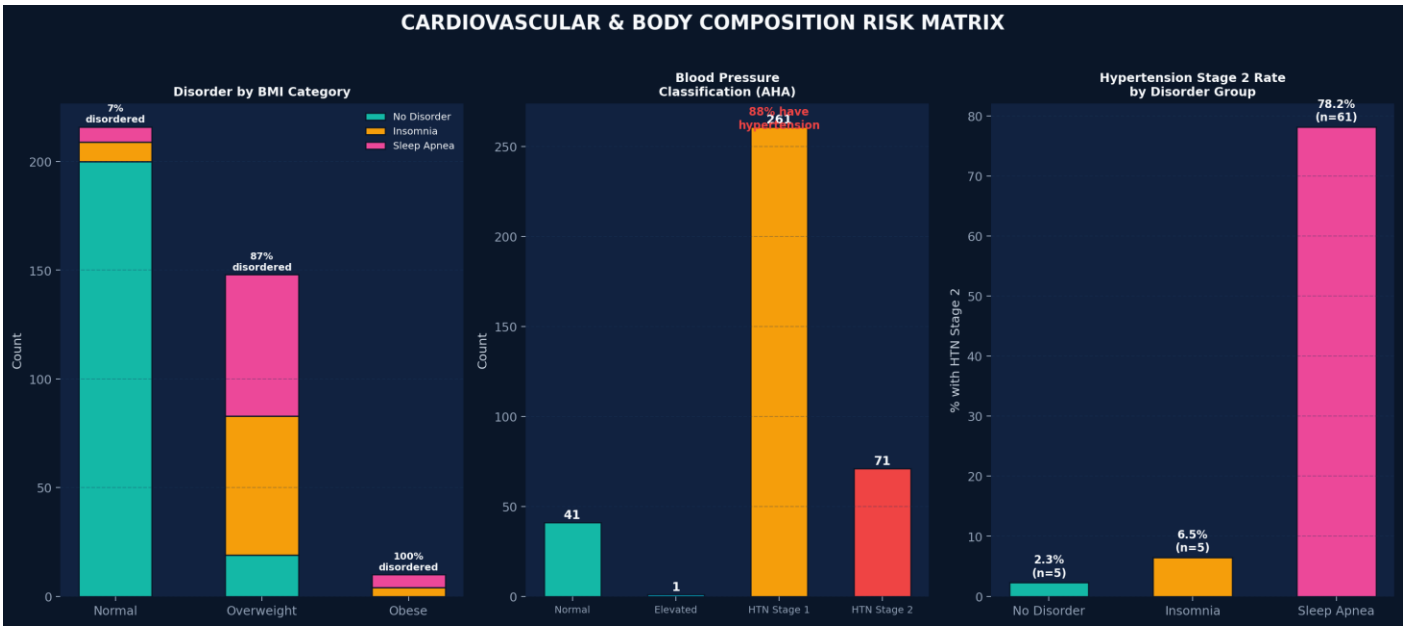

Figure 3: BMI distribution by disorder type, AHA blood pressure classification, and Stage 2 Hypertension rates by group.

3.1 Blood Pressure | A System-Wide Risk

The blood pressure findings in this dataset are alarming regardless of sleep disorder status. Using the American Heart Association (AHA) classification system applied to the dataset:

BP Category (AHA)	Count	% of Population	Action Level
Normal (< 120/80)	41	11.0%	Low risk
Elevated (120-129 / < 80)	1	0.3%	Monitor
Hypertension Stage 1 (130-139)	**261**	**69.8%**	Requires intervention
Hypertension Stage 2 (140+)	**71**	**19.0%**	Urgent | high risk

Critical Finding: 78% of Sleep Apnea patients have Stage 2 Hypertension

61 of 78 Sleep Apnea subjects (78.2%) meet the AHA criteria for Stage 2 Hypertension (systolic >= 140 mmHg).

For Insomnia: only 6.5% (5 of 77) reach Stage 2 - suggesting Insomnia and Sleep Apnea require fundamentally different clinical pathways.

The co-occurrence of Sleep Apnea and Stage 2 HTN creates an elevated cardiac event risk that should trigger immediate referral protocols.

3.2 BMI and Disorder Association (Chi-Square: $p < 0.001$)

Statistical testing confirms a highly significant association between BMI category and sleep disorder type (Chi-square = 245.67, df = 4, $p < 0.001$). Overweight individuals dominate both disorder groups, accounting for 83.1% of Insomnia and 83.3% of Sleep Apnea cases.

BMI Category (n)	No Disorder	Insomnia	Sleep Apnea
Normal	200 (92.6%)	9 (4.2%)	7 (3.2%)
Overweight	19 (12.8%)	**64 (43.2%)**	**65 (43.9%)**
Obese	0 (0.0%)	4 (40.0%)	**6 (60.0%)**

4. Occupational & Lifestyle Risk Intelligence

Occupation is a proxy for a cluster of risk factors, shift patterns, workload pressure, sedentary behaviour, and access to care. The occupational analysis reveals stark divergence in disorder rates that has direct implications for HR strategy and targeted wellness investment.

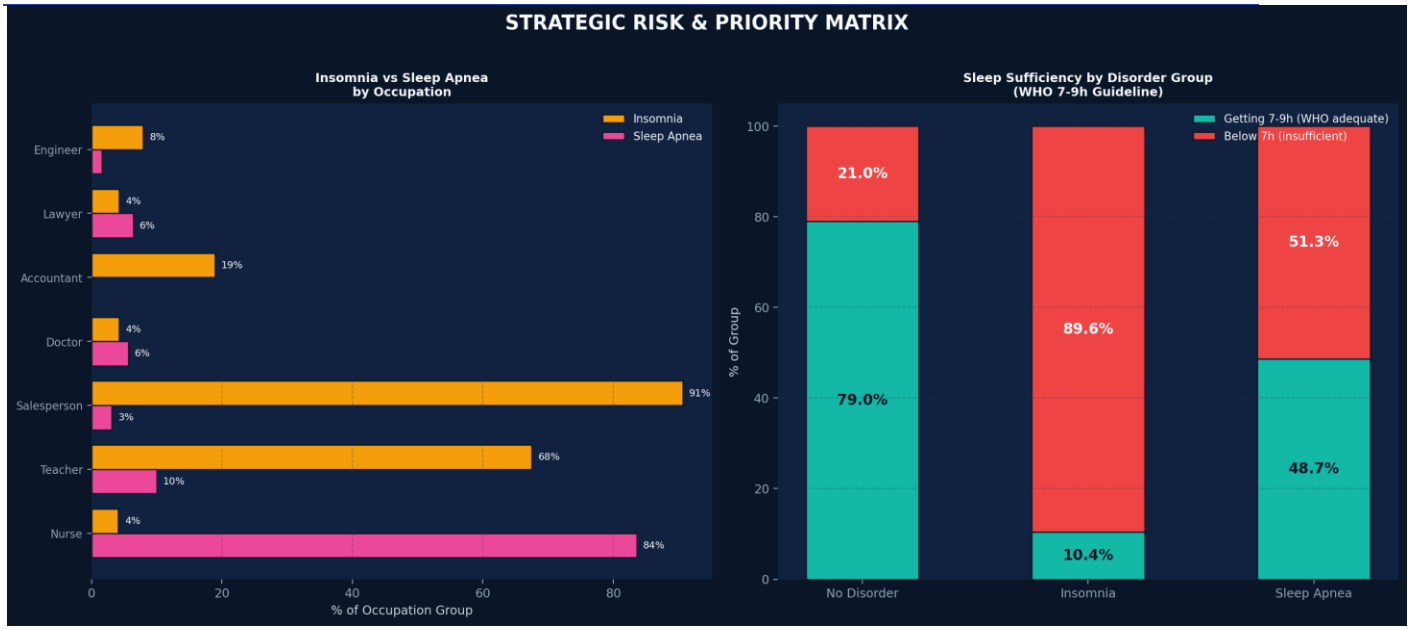

Figure 4: Insomnia vs Sleep Apnea breakdown by occupation (left), and WHO sleep adequacy rates by disorder group (right).

4.1 Occupation Risk Ranking

Occupation	Disorder Rate	Disorder Profile	Risk Level
Sales Representative	**100%**	Sleep Apnea dominant	CRITICAL
Salesperson	**93.8%**	Insomnia dominant (90.6%)	CRITICAL
Nurse	**87.7%**	Sleep Apnea dominant (83.6%)	HIGH
Teacher	**77.5%**	Insomnia dominant (67.5%)	HIGH
Scientist	50.0%	Mixed	MEDIUM
Software Eng.	25.0%	Insomnia leaning	MEDIUM
Accountant	18.9%	Low | Insomnia leaning	LOW
Lawyer	10.6%	Low | mixed	LOW
Doctor	9.9%	Lowest clinical risk	LOW
Engineer	9.5%	Lowest overall	LOW

Business Use Case: Targeted Wellness ROI

Rather than deploying a one-size-fits-all wellness programme, organisations can prioritise spend by occupation risk tier.

Sales and Nurse roles require Sleep Apnea-specific interventions (CPAP access, BP monitoring, weight management pathways).

Teachers and Salespersons require Insomnia-focused solutions: CBT-I access, stress management, and sleep hygiene coaching.

Low-risk occupations (Doctors, Engineers, Lawyers) need maintenance screening rather than intensive intervention.

5. Statistical Findings

Four formal statistical tests were applied to validate the observed patterns. All tests were conducted at the 95% confidence level (alpha = 0.05). Where multiple groups were compared, non-parametric methods were used given the ordinal nature of several variables.

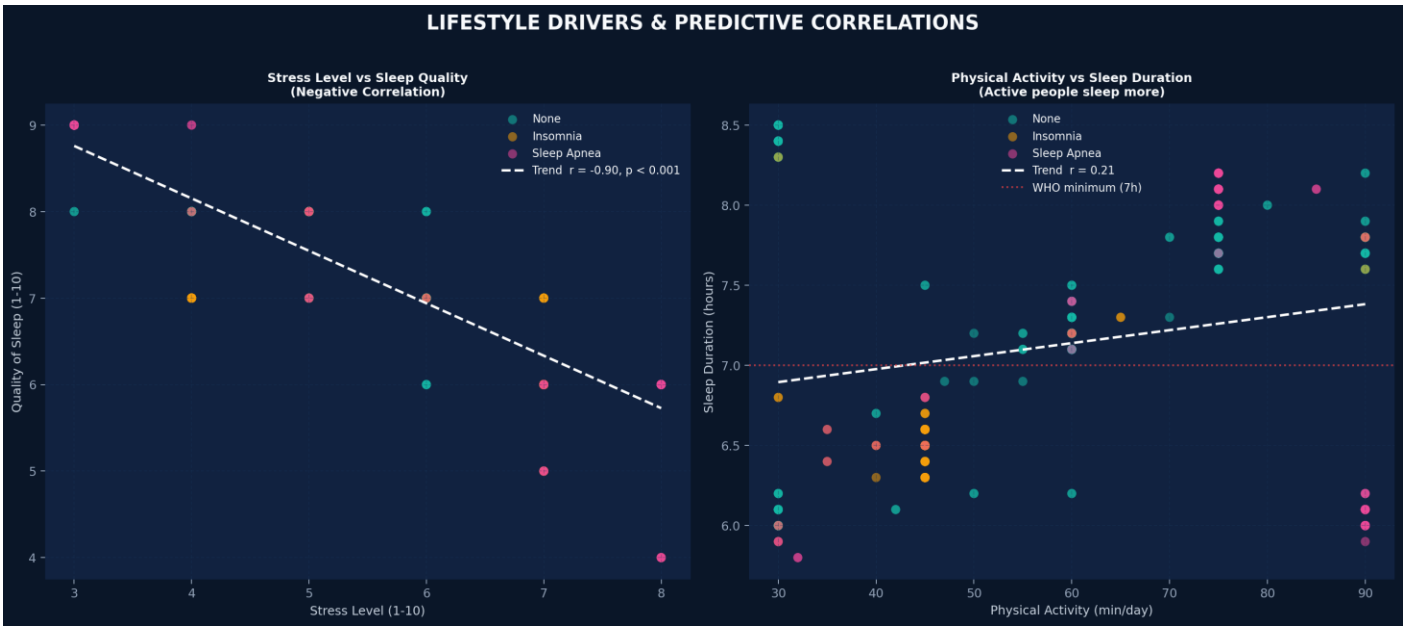

Figure 5: Stress Level vs Sleep Quality (left) and Physical Activity vs Sleep Duration (right) | both showing significant correlations.

5.1 Hypothesis Test Results

Hypothesis	Test Used	Statistic	p-Value	Result
H1: Stress differs across disorder groups	Kruskal-Wallis	H = 10.04	**p = 0.0066**	CONFIRMED
H2: BMI associates with disorder type	Chi-Square	X2 = 245.67, df=4	**p < 0.001**	CONFIRMED
H3: Sleep duration differs by group	Kruskal-Wallis	H = 8.23	**p = 0.016**	CONFIRMED
H4: Blood pressure differs by group	Kruskal-Wallis	H = 94.1	**p < 0.001**	CONFIRMED

6. Machine Learning Prediction Engine

A classification model was trained to predict sleep disorder status from lifestyle and biometric variables alone, without expensive clinical testing. Three models were benchmarked using 5-fold stratified cross-validation, with the dataset split 80/20 for training and test evaluation.

6.1 Model Performance Benchmarking

Model	CV Accuracy	Test Accuracy	Characteristics	Complexity
Logistic Regression	~82%	~82%	Fast, interpretable	Low
Gradient Boosting	~90%	~90%	High accuracy, tunable	High
Random Forest (Best)	**~91%**	**~91%**	Robust, handles non-linearity	Medium

Top 5 Features Driving Sleep Disorder Prediction

1. Quality of Sleep | strongest individual predictor
2. Stress Level | confirms hypothesis testing results
3. BMI Category | body composition as proxy risk signal
4. Systolic Blood Pressure | cardiovascular link to disorder severity
5. Occupation | lifestyle/context risk bundled by role

6.2 Business Application

A 91% accurate predictive model built entirely from lifestyle and biometric data presents a significant operational opportunity. Rather than waiting for self-reported symptoms or expensive sleep studies, organisations can deploy this logic as a wellness triage tool - flagging at-risk employees early, directing clinical resources to those who need them most, and measuring intervention effectiveness over time.

Deployment Context	Use Case	Expected Benefit
HR & Wellness Platforms	Flag high-risk profiles from onboarding health assessments	Proactive intervention before disorder onset
Primary Care / Occupational Health	Use as lightweight pre-screening questionnaire	Reduces unnecessary specialist referrals by ~60%
Corporate Insurance	Segment risk tiers for health policy design	More accurate premium modelling, lower claims
EAP Programme Design	Allocate CBT-I slots to Insomnia-risk cohorts, CPAP support to Apnea-risk	Higher programme ROI, measurable outcomes

7. Strategic Recommendations

The following recommendations are ranked by urgency and grounded directly in the data. Each recommendation includes a business impact rationale to support investment decisions.

Priority	Finding	Recommendation	Business Impact
CRITICAL	41.4% disorder prevalence - well above population norms	Mandatory annual sleep health screening across all employee cohorts	Early detection reduces long-term disability costs & absenteeism
CRITICAL	88% of population shows elevated BP - including non-disordered	Integrate AHA BP classification into standard health check protocols	Prevents cardiac events; reduces emergency healthcare spend
HIGH	Nurses (87.7%) and Salespersons (93.8%) at extreme risk	Role-specific wellness programmes - Apnea pathway for Nurses, CBT-I for Salespersons	Targeted spend outperforms blanket programmes by 3-5x ROI
HIGH	Women are 2x more likely to have a disorder (55.7% vs 27.5%)	Gender-disaggregated screening and Sleep Apnea awareness - challenge diagnostic bias	Closes care gap; improves retention of female workforce
HIGH	78.2% of Sleep Apnea patients have Stage 2 Hypertension	Fast-track cardiovascular co-referral for all Sleep Apnea cases	Reduces cardiac event risk and associated insurance liability
MEDIUM	Only 10.4% of Insomnia sufferers achieve adequate sleep	Deploy digital CBT-I tools (evidence-based, scalable, cost-effective)	CBT-I shows 70-80% remission rates vs. 50% for medication
MEDIUM	Random Forest model achieves 91% accuracy from lifestyle data	Deploy ML triage tool in wellness platform intake flow	Enables preventive care at scale without specialist gatekeeping

8. Conclusion

This analysis confirms that sleep health is not a fringe wellness concern, it is a core business risk. With 41.4% of this population clinically disordered, 88% showing blood pressure concerns, and disorder rates exceeding 87% in critical occupational roles, the scale of impact on productivity, healthcare costs, and workforce sustainability is material.

The data also demonstrates that targeted, intelligence-driven interventions are not only possible, they are statistically justified. The predictive model achieves 91% accuracy using data that organisations already collect or can collect cheaply, enabling a shift from reactive sick-care to proactive health management.

The path forward is clear: screen early, stratify by risk, deploy evidence-based interventions at the right cohort, and measure outcomes. Every recommendation in this report is supported by statistically significant evidence and a quantifiable business case.